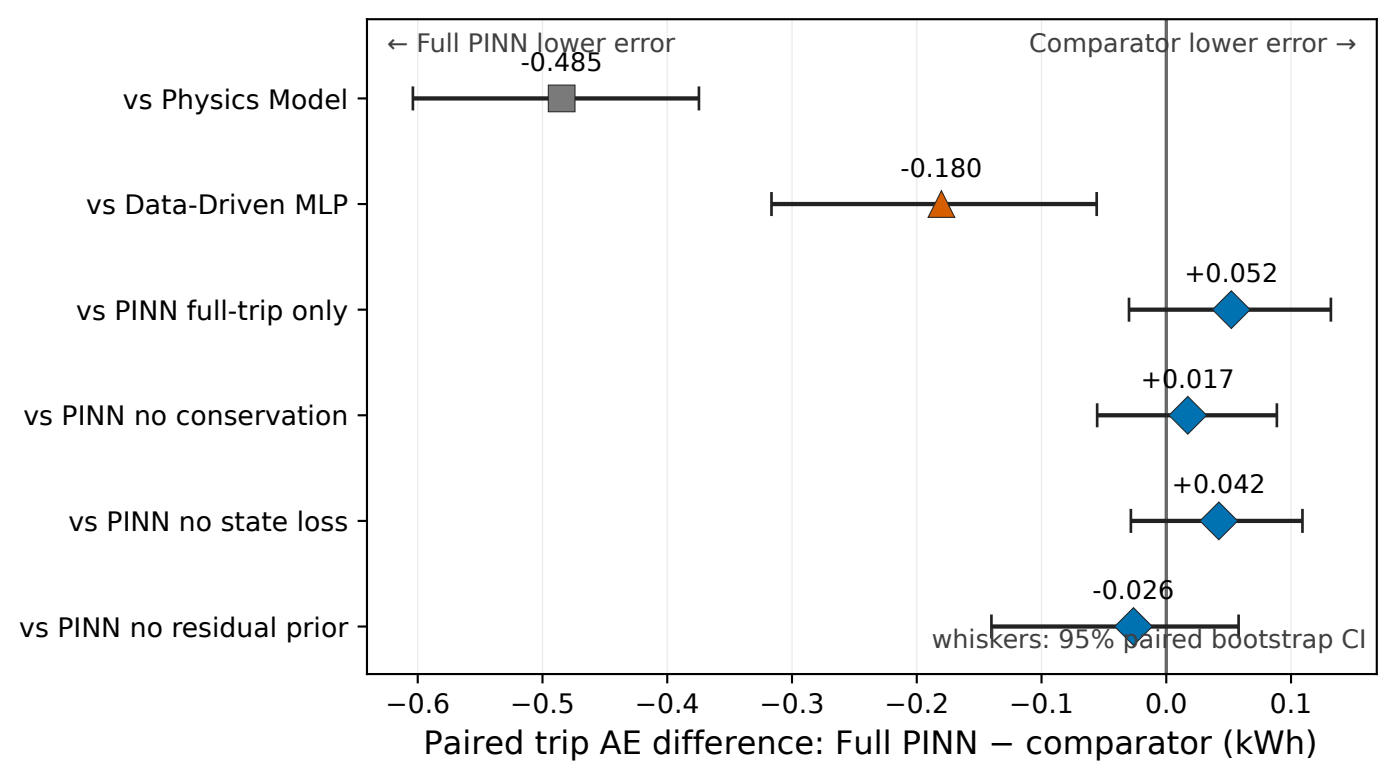
← Full PINN lower error
Comparator lower error →
-0.485
-0.180
+0.052
+0.017
+0.042
-0.026
vs Physics Model
vs Data-Driven MLP
vs PINN full-trip only
vs PINN no conservation
vs PINN no state loss
vs PINN no residual prior
whiskers: 95% paired bootstrap CI
−0.6
−0.5
−0.4
−0.3
−0.2
−0.1
0.0
0.1
Paired trip AE difference: Full PINN − comparator (kWh)